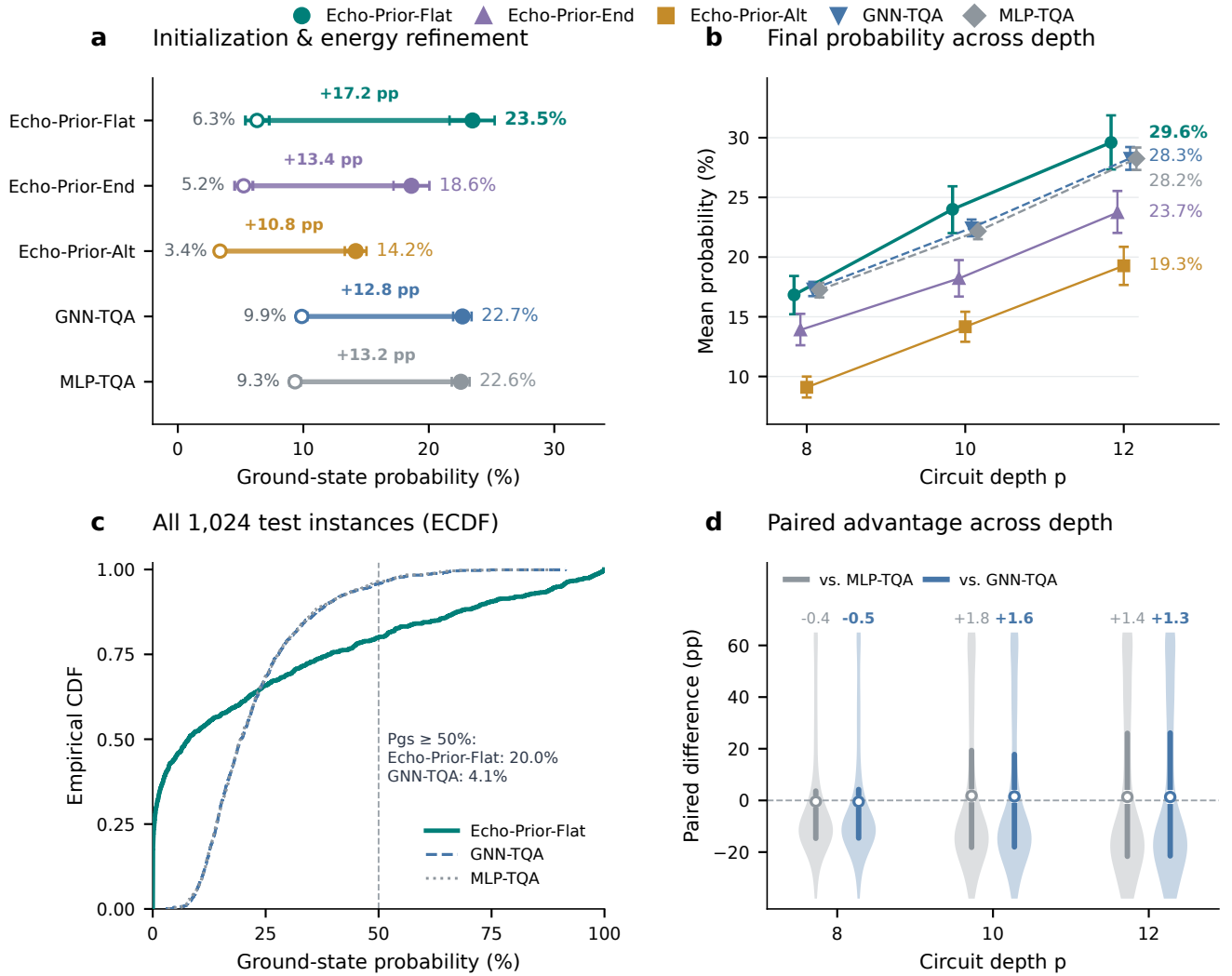

Echo-Prior-Flat
Echo-Prior-End
Echo-Prior-Alt
GNN-TQA
MLP-TQA
a
Initialization & energy refinement
+17.2 pp
6.3%
23.5%
+13.4 pp
5.2%
18.6%
+10.8 pp
3.4%
14.2%
+12.8 pp
9.9%
22.7%
+13.2 pp
9.3%
22.6%
Ground-state probability (%)
b
Final probability across depth
Mean probability (%)
29.6%
28.3%
28.2%
23.7%
19.3%
Circuit depth p
c
All 1,024 test instances (ECDF)
Empirical CDF
Pgs ≥ 50%:
Echo-Prior-Flat: 20.0%
GNN-TQA: 4.1%
Ground-state probability (%)
d
Paired advantage across depth
vs. MLP-TQA
vs. GNN-TQA
-0.4
-0.5
+1.8
+1.6
+1.4
+1.3
Paired difference (pp)
Circuit depth p